Physics - Graduate Level Assessment 9 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. What is the total binding energy and the binding energy per nucleon of the nucleus of _24 $Cr^5$2 which has a mass of 51.95699 amu?
   (a) 4.0×10^8 eV and 7.7×10^6 eV
   (b) 4.2×10^8 eV and 8.1×10^6 eV
   (c) 4.6×10^8 eV and 8.9×10^7 eV
   (d) 4.6×10^8 eV and 8.9×10^6 eV
   (e) 5.0×10^8 eV and 9.6×10^6 eV

2. A car covers a distance of 30 miles in 1/2 hour. What is its speed in miles per hour and in feet per second?
   (a) 30 mi per hr, 60 ft per sec
   (b) 90 mi per hr, 132 ft per sec
   (c) 75 mi per hr, 110 ft per sec
   (d) 60 mi per hr, 55 ft per sec
   (e) 45 mi per hr, 88 ft per sec

3. An object of mass 100 g is at rest. A net force of 2000 dynes is applied for 10 sec. What is the final velocity? How far will the object have moved in the 10-sec interval?
   (a) 150 cm/sec and 7.5 m
   (b) 350 cm/sec and 17.5 m
   (c) 450 cm/sec and 22.5 m
   (d) 250 cm/sec and 12.5 m
   (e) 200 cm/sec and 10 m

4. What is the ratio of intensities of Rayleigh scattering for the two hydrogen lines λ_1 = 410 nm and λ_2 = 656 nm?
   (a) 4.23:1
   (b) 10.24:1
   (c) 8.67:1
   (d) 2.98:1
   (e) 5.67:1

5. Why can't we see the Sun's corona except during total solar eclipses?

(a) We can't see magnetic fields

(b) The corona is blocked by the Earth's atmosphere

(c) The corona is hidden by the Sun's photosphere

(d) The corona is made up mostly of charged particles which aren't luminous.

(e) The Sun's corona is actually invisible to the human eye

True / False (10 marks)

Answer True or False

6. True or False: Bending a branch by hanging on its end causes the top side to experience compressive stress rather than tensile stress.

7. True or False: At 300 K, the wavelength required to promote an electron in germanium is 1600.0 nm, which is within the infrared range.

8. True or False: The period of revolution for two double stars of solar mass separated by 4 light years is approximately 110^8 years.

9. True or False: An ocular lens with a focal length of 4 cm will produce an overall magnification of 100 x in a compound microscope with a 10 x objective.

10. True or False: In a longitudinal wave, the vibrations move in a direction that is along and parallel to the wave's propagation.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Calculate the specific gravity of a rectangular bar of iron with a mass of 320 grams and dimensions of $2 \times 2 \times 10$ cm. Discuss the significance of your findings in the context of material properties.

12. Calculate the rotational kinetic energy of a flywheel with a mass of 30 kg and a radius of gyration of 2 m, rotating at 2.4 revolutions per second. Discuss your calculations and any assumptions made.

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